# Hariprashad Ravikumar

**Machine Learning / Research Engineer** | PhD Physics, Dec 2026 | CUDA · HPC · LLM Systems
San Jose, CA · Available Jan 2027
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## Experience

1. **HAMR Modeling and Simulation Intern**, Western Digital, San Jose, CA *(May 2026 – Aug 2026)*
   - Shipped a full-stack physics simulation web app (Python, Dash, Plotly) that lets engineers run experiments computationally instead of on physical hardware; used daily in production by 40+ senior R&D engineers and subject-matter experts across the company's US and Japan sites
   - Cut per-experiment turnaround for those 40+ engineers with real-time parameter sweeps, side-by-side run comparison, one-click PowerPoint/Excel export, and IndexedDB cross-tab persistence
   - Packaged the tool as a PyPI-installable Python library, deployed it on Kubernetes through a Jenkins CI/CD pipeline, and integrated it into the internal engineering platform alongside related analysis tools

   - Derived a closed-form model of magnetic grain switching from first principles, combining thermal-activation and coherent-rotation theory to predict switching time, signal noise, and error probability across repeated write cycles
   - Validated that model against Monte Carlo simulation and experimental lab data; core findings are in preparation for peer-reviewed publication
   - Identified causal drivers of storage performance through statistical modeling and curve fitting, which let engineers predict and reduce data-integrity errors computationally and removed a hardware-test step from the R&D loop
   - Presented results at PhD Expo 2026 (50+ attendees) and multiple internal engineering reviews (30+ attendees each), including a cross-site review with the company's Japan R&D team
2. **Graduate Research Assistant**, NM State University, Las Cruces, NM *(Aug 2021 – Present)*

   **PhD Project: Lattice Quantum Chromodynamics & Machine Learning Approaches**
   - Generated 30,000+ synthetic data points by solving Partial Differential Equations with Hybrid Monte Carlo simulations (Markov Chain stochastic process), then built an AI for Science pipeline that models the underlying physics to over 98% predictive accuracy using symbolic regression
   - Fit non-linear, high-dimensional models to those 30,000+ points with parallelized Bayesian inference across multi-CPU architectures, using automatic differentiation, full covariance matrices, and jackknife resampling for exact error propagation

## Technical Projects

1. **Cost-Aware Multi-Agent LLM Router** GitHub | Live Demo
   - Cut LLM inference cost 93.9% versus always calling the top-tier model, with no statistically distinguishable accuracy loss on a 50-prompt disjoint holdout
   - Built the routing logic as a calibrated confidence model (logistic regression over logprob uncertainty, self-consistency dispersion, and embedding-distance features) that escalates only the prompts likely to need a costlier model
   - Validated router confidence for honesty as well as raw accuracy using Expected Calibration Error and Brier score, then deployed it as a FastAPI service with Postgres decision logging, Redis caching, and an async circuit breaker on Google Cloud Run
2. **AI-DataScience-Lab: Full-Stack Data Product** GitHub | Live App
   - Shipped a public Flask/Azure statistics app that takes a user from raw upload through Pandas cleaning, Scikit-learn regression, and plotted output in one pass, with an OpenAI-generated narrative summary of the results

## Research Collaborations

1. **Los Alamos National Laboratory** - Lattice Quantum Chromodynamics *(May 2024 – Present)*
    - Ported multi-terabyte Hamiltonian Monte Carlo calculations to parallelized C++/CUDA kernels on GPU-accelerated HPC clusters (NERSC Perlmutter) to cut per-job runtime
    - Ran 75,000+ CPU/GPU compute hours through custom SLURM workflows built for job orchestration at that scale
2. **North Carolina State University** - Mathematical Physics *(Dec 2020 – Present)*
    - Analyzed algebraic symmetry constraints with Mathematica symbolic computation on HPC clusters, producing a Physical Review D paper publication.

## Selected Publications

Google Scholar

1. C.-R. Ji and **H. Ravikumar**, *"Interpolating conformal algebra in (1+1) dimensions,"* **Physical Review D** 113, 096018 (May 2026). DOI: 10.1103/prlq-4j1l

## Technical Skills

**Methods & HPC:** Parallel Computing (GPU, MPI), Numerical Methods (PDEs, Monte Carlo, Regression)
**Programming:** Python, C++, CUDA, Bash, SQL
**ML & APIs:** PyTorch, TensorFlow, Scikit-learn, Pandas, Dash, Plotly, Flask, FastAPI, Numba
**Cloud & MLOps:** Azure, AWS (S3, Lambda), Docker, Kubernetes, Jenkins, CI/CD, Git, SLURM

## Education

**PhD in Physics**, New Mexico State University, USA *Aug 2021 – Dec 2026 (expected)*
**MS in Physics**, New Mexico State University, USA *Aug 2021 – May 2024*

## Awards

- **2025 NMC Collaboration Grant**, awarded by the New Mexico Consortium for an independent research project in collaboration with scientists at Los Alamos National Laboratory
- **2023 George and Barbara Goedecke Physics Excellence Fund Scholarship**, awarded by the NMSU Physics Department

## Leadership

- **Vice President**, Physics Graduate Student Organization (NMSU) *Aug 2025 – Present*
  Organized professional development events and served as primary liaison between 40+ graduate students and faculty